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THE HONG KONG
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ELEC 3200: System Modeling, Analysis and Control

Lecture 11: Modeling of Closed-Loop Systems

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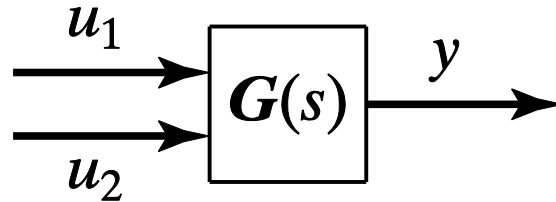
MISO and SIMO Systems

- So far we have dealt with mostly SISO systems, but a general system might be MIMO.
- We will occasionally deal with **MISO systems** or **SIMO systems**.
 - MISO systems and SIMO systems are frequently seen in practice and they can be dealt with using techniques not much beyond the theory for SISO systems.
- In particular, in the control of SISO systems, such as in the regulation control, we often need to use MISO systems, leading to better performance.
- Without loss of generality, we will study only **double-input-single-output (DISO)** systems and **single-input-double-output (SIDO)** systems.

DISO Systems

- A system with two inputs and one output
- The transfer function and block diagram

$$\mathbf{G}(s) = [G_1(s) \quad G_2(s)]$$

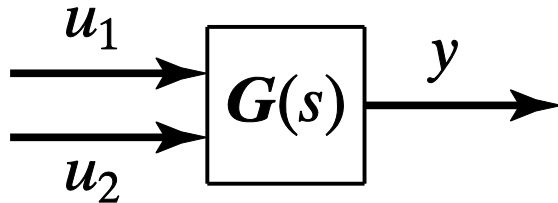


- Input/output relation

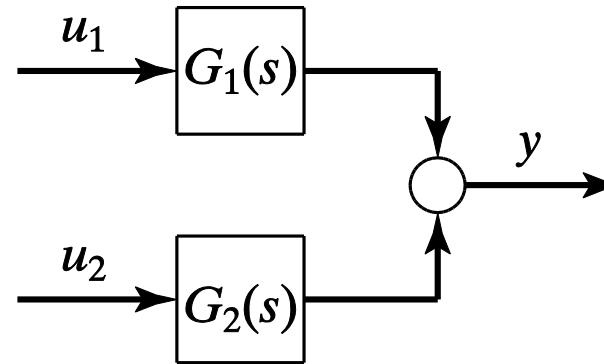
$$Y(s) = [G_1(s) \quad G_2(s)] \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix} = G_1(s)U_1(s) + G_2(s)U_2(s).$$

DISO Systems

- A DISO system



- A **misleading** view



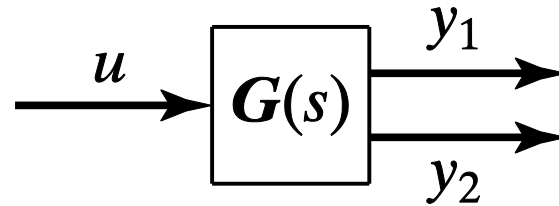
- These two block diagrams are only **algebraically equivalent**.
- Better to view a DISO system as an **inseparable single system**.

$$\begin{aligned} G(s) &= \frac{\mathbf{b}(s)}{a(s)} = \frac{1}{a(s)} \begin{bmatrix} b_1(s) & b_2(s) \end{bmatrix} \\ &= \frac{\begin{bmatrix} b_{10}s^n + \cdots + b_{1n} & b_{20}s^n + \cdots + b_{2n} \end{bmatrix}}{a_0s^n + a_1s^{n-1} + \cdots + a_n}, \quad a_0 \neq 0. \end{aligned}$$

SIDO Systems

- A system with one input and two outputs
- The transfer function and block diagram

$$\mathbf{G}(s) = \begin{bmatrix} G_1(s) \\ G_2(s) \end{bmatrix}$$



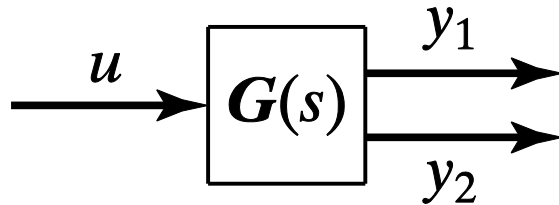
- Input/output relation

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} G_1(s) \\ G_2(s) \end{bmatrix} U(s),$$

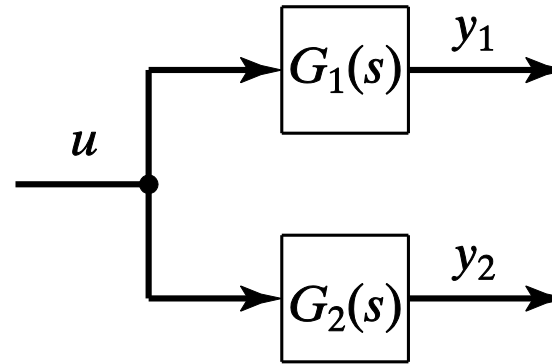
$$\iff Y_1(s) = G_1(s)U(s), \quad Y_2(s) = G_2(s)U(s).$$

SIDO Systems

- A SIDO system



- A **misleading** view

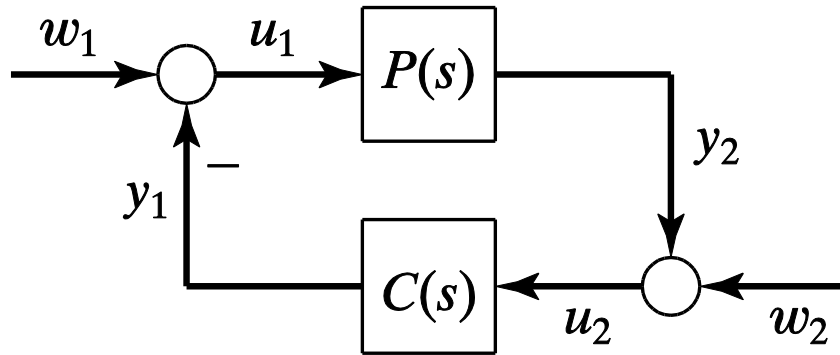


- These two block diagrams are only **algebraically equivalent**.
- Better to view a SIDO system as an **inseparable single system**.

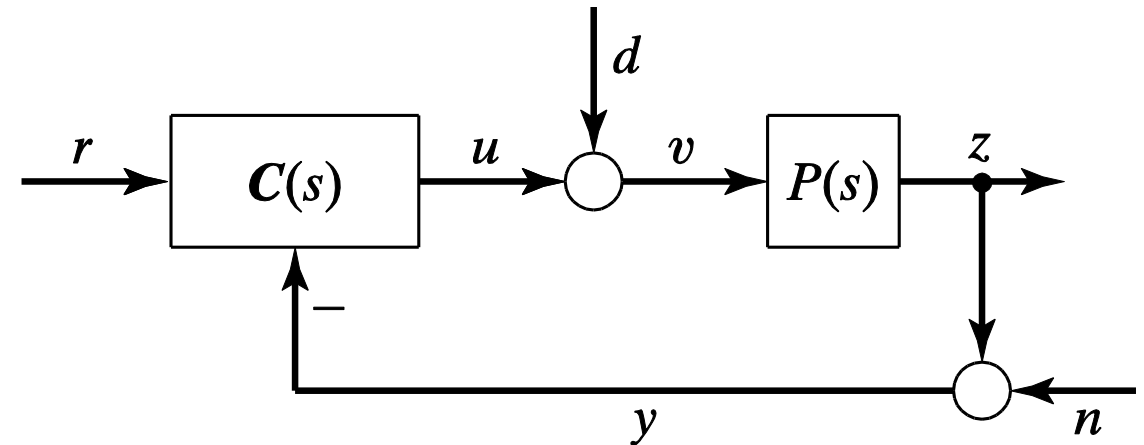
$$\mathbf{G}(s) = \frac{\mathbf{b}(s)}{a(s)} = \frac{1}{a(s)} \begin{bmatrix} b_1(s) \\ b_2(s) \end{bmatrix} = \frac{\begin{bmatrix} b_{10}s^n + \dots + b_{1n} \\ b_{20}s^n + \dots + b_{2n} \end{bmatrix}}{a_0s^n + a_1s^{n-1} + \dots + a_n}, \quad a_0 \neq 0.$$

Modeling of Closed-Loop systems

- Feedback system for stabilization



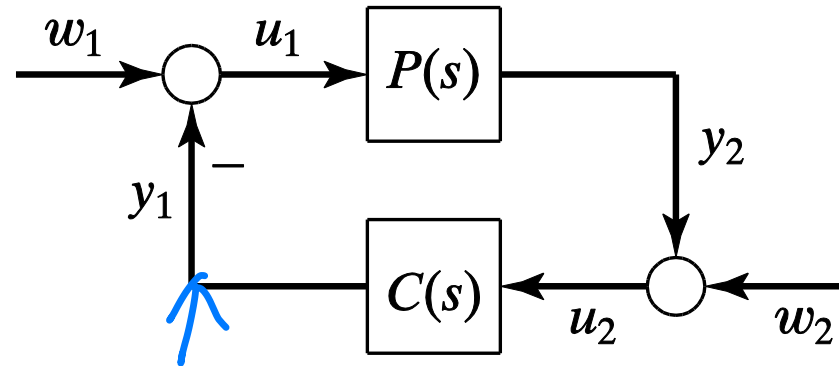
- Feedback system for regulation



- **Stabilization problem:** Given plant P , design controller C so that the system has **good stability**.
- **Regulation problem:** Given plant P , design two-degree-of-freedom (2DOF) controller $C = \begin{bmatrix} C_1 & C_2 \end{bmatrix}$ so that **good performance** in **tracking** and **disturbance rejection** is achieved.

Feedback System for Stabilization

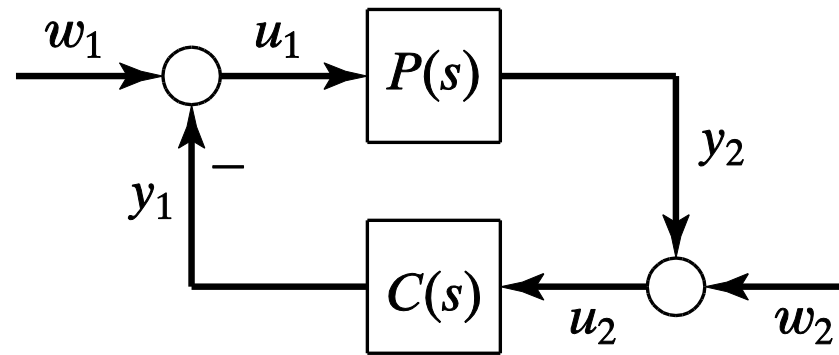
- Well posed feedback systems for stabilization



- The transfer function from w_1 to y_1 is $\frac{P(s)C(s)}{1 + P(s)C(s)}$.
- In order for the transfer function to be meaningful, $1 + P(s)C(s) \neq 0$.
- **Definition** The feedback system for stabilization is said to be **well posed** if $1 + P(s)C(s) \neq 0$. Otherwise, it is said to be **ill posed**.

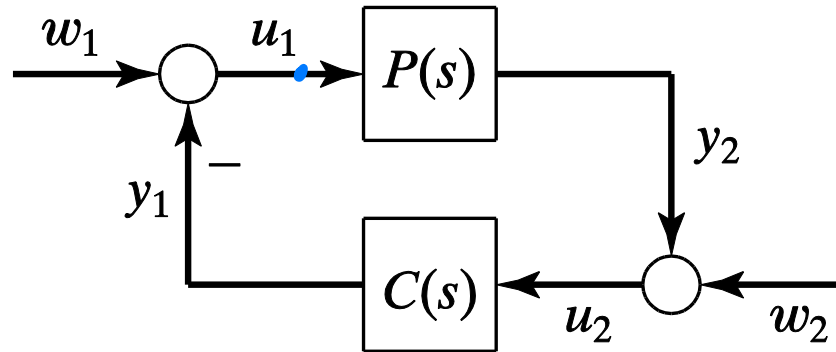
Feedback System for Stabilization

► Example



- If $P(s) = 1$ and $C(s) = -1$, then the system is **ill posed**. In this case, the internal signals cannot be uniquely determined from the external signals.

Feedback System for Stabilization

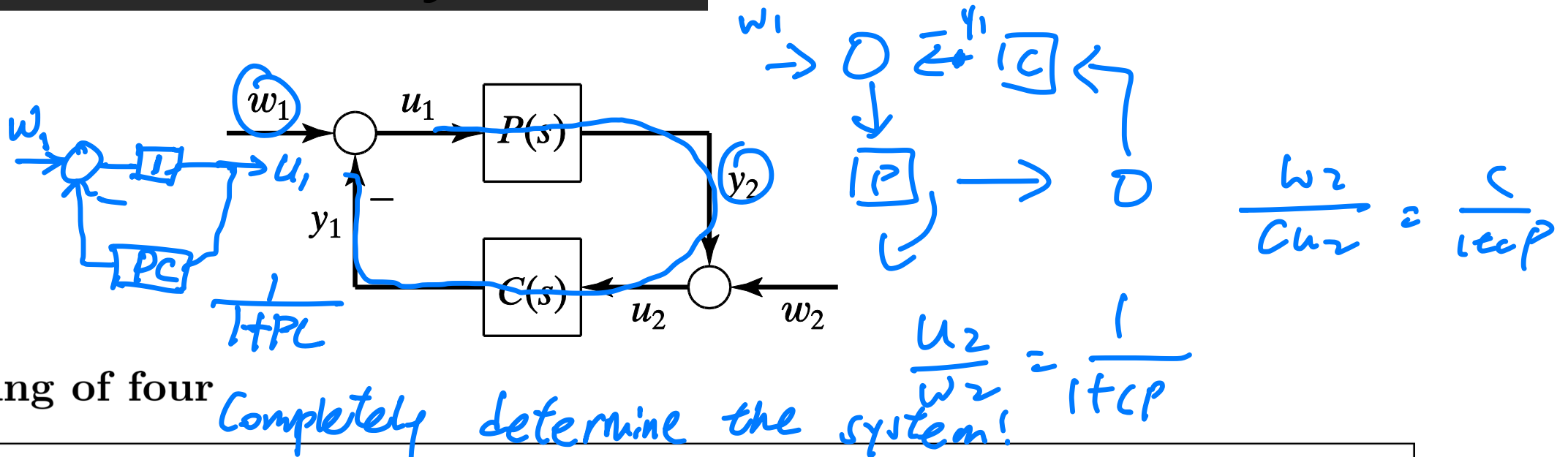


$$\frac{Y_2}{W_1} = \frac{P}{1+CP}$$

- All the transfer functions from one of the external signals to one of the internal signals are

$$\begin{bmatrix} U_1(s) \\ U_2(s) \\ Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} \frac{1}{1 + P(s)C(s)} & \frac{-C(s)}{1 + P(s)C(s)} \\ \frac{P(s)}{1 + P(s)C(s)} & \frac{1}{1 + P(s)C(s)} \\ \frac{P(s)C(s)}{1 + P(s)C(s)} & \frac{C(s)}{1 + P(s)C(s)} \\ \frac{P(s)}{1 + P(s)C(s)} & \frac{-P(s)C(s)}{1 + P(s)C(s)} \end{bmatrix} \begin{bmatrix} W_1(s) \\ W_2(s) \end{bmatrix}$$

Feedback System for Stabilization



- Gang of four

Completely determine the system!

$\frac{1}{1 + P(s)C(s)}$	$\frac{C(s)}{1 + P(s)C(s)}$	$\frac{P(s)}{1 + P(s)C(s)}$	$\frac{P(s)C(s)}{1 + P(s)C(s)}$
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$$\frac{w_1}{y_2} = \frac{1}{1 + PC}$$

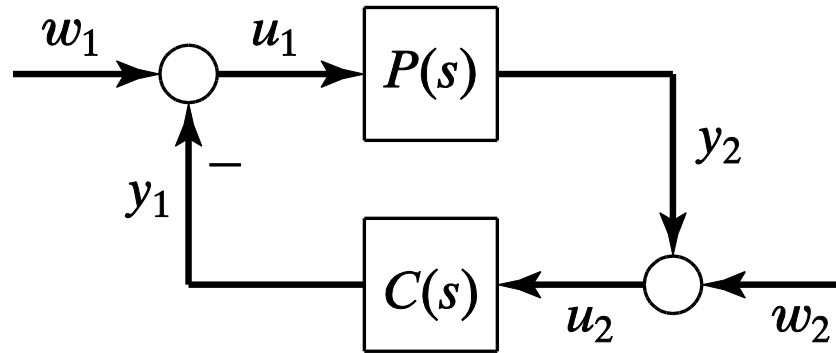
$$\frac{w_1}{pu_1} = \frac{1}{1 + PC}$$

$$\Rightarrow \frac{w_1}{u_1} = \frac{P}{1 + PC}$$

$$\frac{w_2}{y_1} = \frac{C}{1 + PC}$$

$$\frac{w_2}{C u_2} = \frac{C}{1 + PC}$$

Feedback System for Stabilization



- **Sensitivity function** $S(s)$ and **complementary sensitivity function** $T(s)$

$$S(s) = \frac{1}{1 + P(s)C(s)} \quad \text{and} \quad T(s) = \frac{P(s)C(s)}{1 + P(s)C(s)},$$

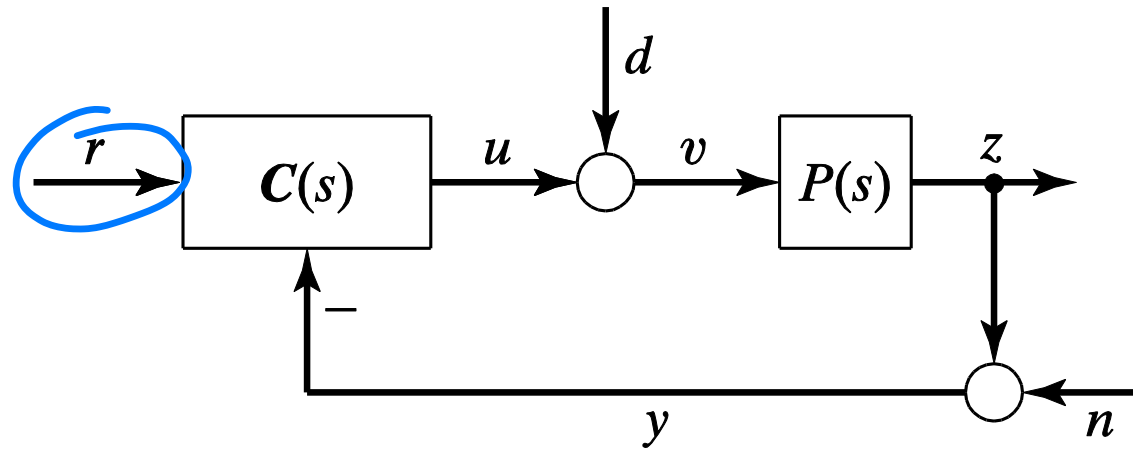
$$S(s) + T(s) = 1.$$

- **Alternative expressions of gang of four**

$$S(s), \quad C(s)S(s), \quad P(s)S(s), \quad T(s).$$

Feedback System for Regulation

- 2DOF regulation control systems



Give another variables

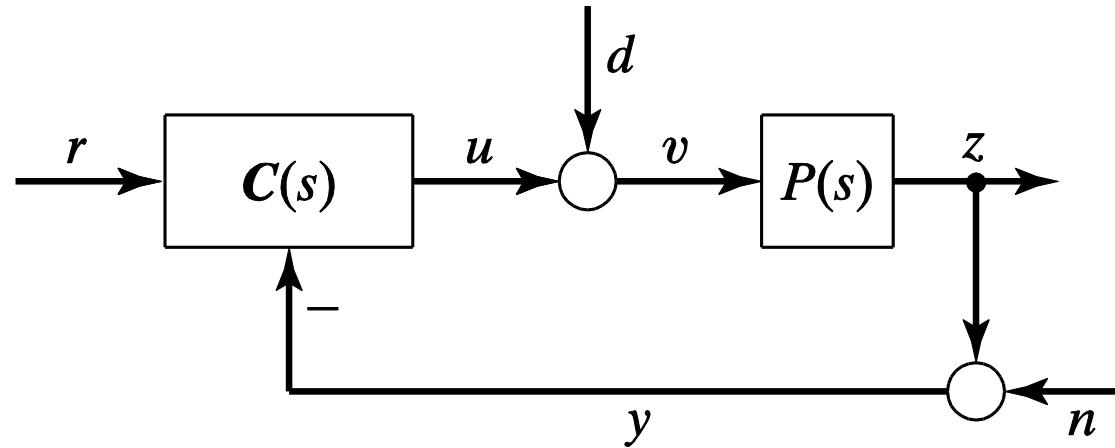
- $C(s)$ is a **two-degree-of-freedom**, or simply a **2DOF**, controller

$$U(s) = \begin{bmatrix} C_1(s) & -C_2(s) \end{bmatrix} \begin{bmatrix} R(s) \\ Y(s) \end{bmatrix}$$

- A minus sign in front of $C_2(s)$ because this suggests a **negative feedback**.

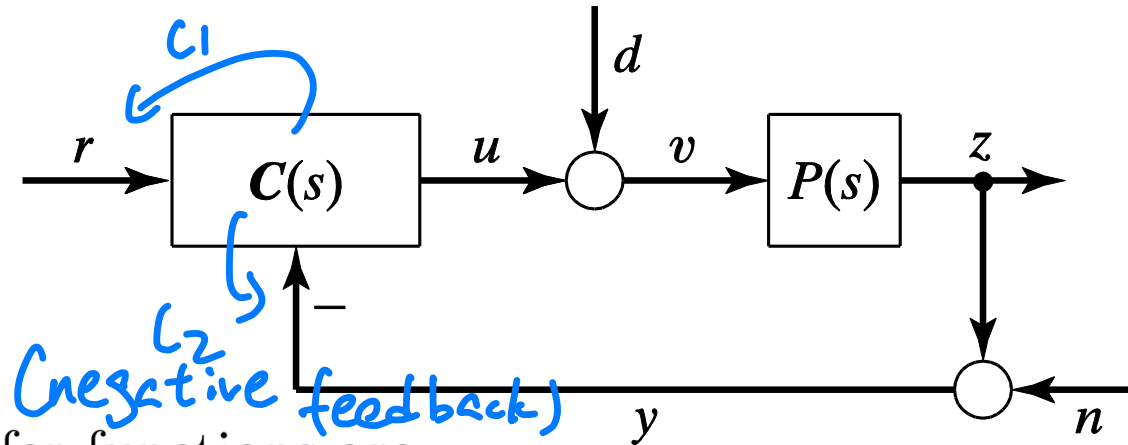
Feedback System for Regulation

- 2DOF regulation control systems



- **Definition** The feedback system for regulation is said to be **well posed** if the feedback system for stabilization formed by $P(s)$ and $C_2(s)$ is **well posed**.

Feedback System for Regulation



$$z = ? N(s)$$

$$\frac{y}{n} = \frac{1}{1+CP}$$

$$z = y - n$$

- All the transfer functions are

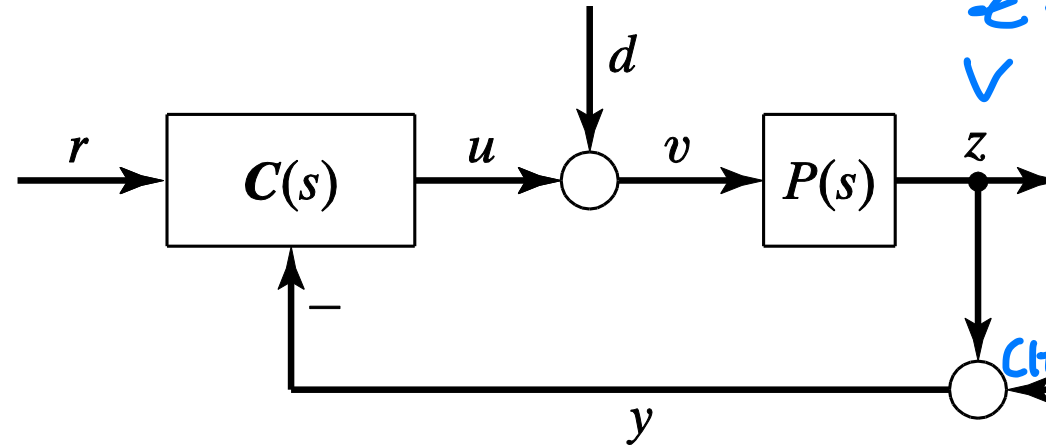
$$\begin{bmatrix} U(s) \\ V(s) \\ Y(s) \\ Z(s) \end{bmatrix} = \begin{bmatrix} \frac{-P(s)C_2(s)}{1 + P(s)C_2(s)} & \frac{-C_2(s)}{1 + P(s)C_2(s)} & \frac{C_1(s)}{1 + P(s)C_2(s)} \\ \frac{1}{1 + P(s)C_2(s)} & \frac{-C_2(s)}{1 + P(s)C_2(s)} & \frac{C_1(s)}{1 + P(s)C_2(s)} \\ \frac{P(s)}{1 + P(s)C_2(s)} & \frac{1}{1 + P(s)C_2(s)} & \frac{P(s)C_1(s)}{1 + P(s)C_2(s)} \\ \frac{P(s)}{1 + P(s)C_2(s)} & \frac{-P(s)C_2(s)}{1 + P(s)C_2(s)} & \frac{P(s)C_1(s)}{1 + P(s)C_2(s)} \end{bmatrix}$$

$$\frac{z+n}{n} = \frac{1}{1+CP}$$

$$z = n \left(\frac{1-1-CP}{1+CP} \right)$$

$$\begin{bmatrix} D(s) \\ N(s) \\ R(s) \end{bmatrix} \frac{z}{n} = \frac{-CP}{1+CP}$$

Feedback System for Regulation



$$z = PV$$

$$v = utd$$

$$u = RC_1 - C_2[nt + P(u+d)]$$

$$u = RC_1 - C_2N - PC_2u - PC_2D$$

$$(1 + PC_2)u = C_1R - C_2N - PC_2D$$

$$u = \frac{-PC_2D + C_1R - C_2N}{1 + PC_2}$$

- **Gang of four** formed by $P(s)$ and $C_2(s)$

$$\frac{1}{1 + P(s)C_2(s)}, \quad \frac{C_2(s)}{1 + P(s)C_2(s)}, \quad \frac{P(s)}{1 + P(s)C_2(s)}, \quad \frac{P(s)C_2(s)}{1 + P(s)C_2(s)}.$$

- Together with **two additional ones** involving $C_1(s)$

$$\frac{C_1(s)}{1 + P(s)C_2(s)}, \quad \frac{P(s)C_1(s)}{1 + P(s)C_2(s)}.$$

Feedback System for Regulation

- Gang of six

$$\begin{array}{ccc} \frac{1}{1 + P(s)C_2(s)}, & \frac{C_2(s)}{1 + P(s)C_2(s)}, & \frac{P(s)}{1 + P(s)C_2(s)}, \\ \frac{P(s)C_2(s)}{1 + P(s)C_2(s)}, & \frac{C_1(s)}{1 + P(s)C_2(s)}, & \frac{P(s)C_1(s)}{1 + P(s)C_2(s)}. \end{array}$$

- **Sensitivity function** $S(s)$ and **complementary sensitivity function** $T(s)$

$$S(s) = \frac{1}{1 + P(s)C_2(s)} \quad \text{and} \quad T(s) = \frac{P(s)C_2(s)}{1 + P(s)C_2(s)}.$$

- Alternative expressions of gang of six

make sure that designing C_1 will be stable!

$$S(s), \quad C_2(s)S(s), \quad P(s)S(s), \quad T(s), \quad C_1(s)S(s), \quad P(s)C_1(s)S(s).$$

Feedback System for Regulation

- Gang of six matrix

$$\begin{aligned} GoS(s) &= \begin{bmatrix} \frac{1}{1 + P(s)C_2(s)} & \frac{C_2(s)}{1 + P(s)C_2(s)} & \frac{C_1(s)}{1 + P(s)C_2(s)} \\ \frac{P(s)}{1 + P(s)C_2(s)} & \frac{P(s)C_2(s)}{1 + P(s)C_2(s)} & \frac{P(s)C_1(s)}{1 + P(s)C_2(s)} \end{bmatrix} \\ &= \begin{bmatrix} S(s) & C_2(s)S(s) & C_1(s)S(s) \\ P(s)S(s) & T(s) & P(s)C_1(s)S(s) \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ P(s) \end{bmatrix} S(s) \begin{bmatrix} 1 & C_2(s) & C_1(s) \end{bmatrix}. \end{aligned}$$

Review & Exercise

- How to represent the DISO and SIDO system?
- What's the definition of “well posed” and “ill posed”?
- Know how to write the transfer function form for the stabilization and regulation.
- Know the “Gang of four” and “Gang of six”.
- Know the “sensitivity function” and “complementary sensitivity function”